Enrollment No._____

GANPAT UNIVERSITY
B.Sc.IT (DATA SCIENCE) SEM-III EXAMINATION

U73A1DS: DATA STRUCTURE

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Write a short note on the Classification of Data Structure.
2. Explain the operations of Stack with an example.
3. What are the Differences between the Deque vs Circular Queue
4. Create algorithm for inserting an element in circular link list.
5. Explain Binary Search Tree with example
6. Describe Types of Searching.
7. Write down the algorithm for the Bubble sort.

SECTION-II

2. Answer the following: (Any two) **(12)**

1. Write an algorithm for inserting an element in a stack, removing an element from stack.
2. Write an algorithm of Simple queue
3. Consider singly linked storage structures, write an algorithm which performs an Deletion at the end

3. Answer the following: (Any two) **(12)**

1. What is Binary tree? Explain Types of Binary tree
2. Illustrate AVL tree with example
3. Explain algorithm of Sequential (Linear) search

4. Answer the following: (Any One) **(06)**

1. Create Program for the Selection Sorting.
2. Explain the Quick Sort with suitable Example.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. IT (Data Science) SEM-III EXAMINATION
U73A2ADS: ADVANCE DATABASE MANAGEMENT SYSTEM

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five)** **(30)**

1. What is recovery? Explain recovery types.
2. Explain The Generic PL/SQL block with example.
3. Write a short note on oracle stored package.
4. What is moving data? Discuss general architecture.
5. Discuss brief about control file with example.
6. Explain function in PL/SQL with example.
7. Explain conditional statement in PL/SQL.

SECTION-II

2. **Answer the following: (Any Two)** **(12)**

1. What is concurrency? Explain concurrency problems with example.
2. What is cursor? Explain implicit cursor with example.
3. What is deadlock? How to control deadlock.

3. **Answer the following: (Any Two)** **(12)**

1. Explain triggers with example.
2. What is data pump? Explain import and export.
3. What is exception? Explain with example.

4. **Answer the following: (Any One)** **(06)**

1. Explain any six SQL* loader commands.
2. What is SQL*loader? Explain SQL*loader datatypes.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. IT(DS) SEM-III EXAMINATION

U73A3OCP: OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five)** (30)

1. What is Object-oriented Programming? Explain the advantages of using OOPs.
2. Explain the use of super and this keyword with a suitable example.
3. Describe the process of importing and accessing a package with suitable examples.
4. Differentiate between a constructor and a method of the class.
5. What are lambda expressions? How to implement it in Java programming language?
6. What is Phaser? Explain in brief.
7. Explain Dynamic Array in Detail.

SECTION-II

2. **Answer the following: (Any Two)** (12)

1. What is an Interface? Give an example where Interface can be used to support multiple inheritance.
2. Demonstrate Abstraction using abstract class and method.
3. Explain the significance of private, protected, and public access specifiers.

3. **Answer the following: (Any Two)** (12)

1. How can we add a class to a package? Write about relative and absolute paths.
2. How are Lambda expression and functional interfaces related?
3. What are Generics? Explain its advantages with examples.

4. **Answer the following: (Any One)** (06)

1. What is Executor? Explain in brief.
2. Explain in detail the fork/join framework in concurrency utilities.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc.IT (DATA SCIENCE) SEM-III EXAMINATION

U73A4AS ADVANCE STATISTICS

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer
4. Scientific calculator is allowed

SECTION-I

1. **Answer the following: (Any Five)** **(30)**

1. Find a positive root of the equation $2sinx - x = 0$ by bisection method correct to two decimal places.
2. If $x = 3.141592$ and approximate value $\bar{x} = 3.14$ then find the following
   (i)Absolute error (ii)Relative error and (iii)Percentage error.
3. Evaluate the unknowns of the system of linear equations
   $5x - 2y + 3z = 18$
   $x + 7y - 3z = -22$
   $2x - y + 6z = 22$
   by Gauss elimination method.
4. A person plays a game of throwing a die under the condition that he could get as many rupees as the number of points on the uppermost face. Find the expectation and variance of his winnings.
5. A random sample of 500 oranges was taken from a large consignment and 65 were found to be defective. Find the standard error of the sample population of defective oranges for samples of size 500.
6. Fit a second degree trend equation from the following data

x:	0	1	2	3	4
y:	1	1.8	1.3	2.5	6.3

7. A can hit the target three times in five shots, B can hit a target two times in five shots, C can hit a target three times in four shots. What is the probability that
   (i)Two shots hit (ii) At least two shots hit.

SECTION-II

2. **Answer the following: (Any Two)** **(12)**

1. Using Newton-Raphson method, find the real root of $x^3 + x - 1 = 0$ correct to three decimal places.
2. Evaluate the unknowns of the system of linear equations
   $2x + 5y - 3z = 1$
   $5x + y + 4z = 2$
   $7x + 3y + z = 4$
   by Gauss-Jordan method
3. Evaluate the unknowns of the system of linear equations
   $2x + y + 2z = 10$
   $x + 2y + z = 8$
   $3x + y - z = 2$
   by Gauss-Jordan method

3. **Answer the following: (Any Two)** **(12)**

1. Discuss the non-probability sampling methods.
2. Discuss the any four types of probability sampling methods.
3. If three coins are tossed find the expectation and the variance of the number of heads.

4. **Answer the following: (Any One)** **(06)**

1. Discuss components of time series analysis.
2. From the following data find the trend values by 5 yearly moving averages

Year:	1970	1971	1972	1973	1974	1975	1976	1977	1978
Sales:	36	43	43	34	44	54	34	24	14

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

B.Sc.I.T. (Data Science) SEM-III EXAMINATION (CBCS)

U73A5AI: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any 5) **[30]**

1. What is AI? Explain types of AI with example.
2. What are some real-life applications of Artificial Intelligence?
3. How to define the problem? Explain state space problem with example.
4. Explain various AI problem characteristics with example.
5. Discuss the production system, and characteristics of production system.
6. What is natural language processing? Discuss the steps in natural language processing.
7. Discuss the Discourse and Pragmatic Processing in detail.

SECTION-II

2. Answer the following: (Any 2) **[12]**

1. What is AI agent? Explain Types of AI agent.
2. Explain distributed reasoning systems in detail.
3. Explain knowledge representation Framework in detail.

3. Answer the following: (Any 2) **[12]**

1. Discuss forward verses backward reasoning with example.
2. What is reasoning? Explain types of reasoning in detail.
3. Explain AI knowledge life cycle.

4. Answer the following: (Any 1) **[06]**

1. Explain Probabilistic reasoning with Bay's Theorem.
2. Discuss issue in the design of search program with example of BFS.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY
B.Sc.IT (Data Science) SEM-III EXAMINATION

U73B6EDM: ENVIRONMENT DISASTER MANAGEMENT

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five out Seven)** **(30)**

1. Discuss Challenges faced by Environment.
2. What is Ecosystem? Explain the major part of ecosystem.
3. Explain different Types of Air Pollutants in brief.
4. Discuss how many ways we control the Noise pollution.
5. Write a short note on Current Environmental Global Issues: - Global Warming.
6. Discuss the Awareness And Planning for Flood Disaster.
7. Explain role of Government in rehabilitation.

SECTION-II

2. **Answer the following: (Any Two)** **(30)** **(12)**

1. What is Environment? Discuss the Scope of environment.
2. Discuss the Structure of atmosphere in detail.
3. Discuss types of ecosystem.

3. **Answer the following: (Any Two)** **(12)**

1. Discuss Ozone Layer Depletion - Cause, Effects, And Solutions.
2. What is Disaster? Write short note on Disaster management Cycle.
3. Discuss the Residential and Industrial Fires Prevention tips in brief.

4. **Answer the following: (Any One)** **(6)**

1. Discuss Role of Local Institutions: - Media in Rehabilitation.
2. Discuss importance of Insurance company as a Local Institutions in Rehabilitation.

END OF PAPER